Enough about probability (for now). Let's turn now to some results from contemporary physics which will be important to the argument which follows.

A first question: what does the theory provided by physics include?

‘‘The standard model of physics presents a theory of the electromagnetic, weak, and strong forces, and a classification of all known elementary particles. The standard model specifies numerous physical laws, but that's not all it does. According to the standard model there are roughly two dozen dimensionless constants that characterize fundamental physical quantities.’’ (Hawthorne & Isaacs, ‘‘Fine-tuning fine-tuning’’)

These “dimensionless constants” will be our focus.

One such constant is the cosmological constant, which measures the energy density of empty space.

The cosmological constant is just a number which (as far as we know) could have different values consistent with the laws of physics.

One thing that the standard model of physics gives us is a measure of how likely it is, given the laws of the nature, that the fundamental constants (like the cosmological constant) would fall in a certain range.

We can make certain plausible assumptions about what it would take for life to have evolved. For example, if there were nothing but hydrogen, it is hard to see how life could have evolved. If there were no planets, it is hard to see how life could have evolved.

Given assumptions such as these, we can look at what the standard model of physics tells us about how likely it is that, for example, the cosmological constant has a value which would permit the evolution of life.

“Physicists have determined the (approximate) values of the fundamental constants by measurement. (There's no way to derive the values of the fundamental constants from other aspects of the standard model. Any quantities that could be so derived wouldn't be fundamental.) Still, the underlying theory favored some sorts of parameter-values over others. ... Physicists made the startling discovery that -- given antecedently plausibly assumptions about the nature of the physical world -- the probability that a universe with general laws like ours would be habitable was staggeringly low.”

This is the claim that that the cosmological constant having a life-supporting value, given the laws of nature, is very low.

We should emphasize just how small we take the life-permitting parameter values to be according to the physically-respectable measures. "Small" here doesn't mean "1 in 10,000" or "1 in 1,000,000". It means the kind of fraction that one would resort to exponents to describe, as in "1 in 10 to the 120". The kind of package that we have in mind tells us that only a fantastically small range is life permitting.

The claim is that, according to current physics, the probability of the cosmological constant falling in a life-supporting range, given the laws of nature, is $\frac{1}{10^{120}}$.

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Let's now ask how to turn these facts about current physics and probability theory into an argument for the existence of God.

Much as we considered two different hypotheses about the number of balls in the urn, so we can consider two different hypotheses about the origins of the universe.

Let the **design hypothesis** be the hypothesis that the universe was created by an intelligent designer. Let the **non-design** hypotheses be the hypothesis that it was not.

Let our **evidence** (e) be the fact that the cosmological constant has a value in the life-permitting range.

Then what do we know about the relevant probabilities?

$$\boxed{Pr(e \mid \text{non-design}) = \frac{1}{10^{120}}}$$

$$Pr(e \,|\, \text{design}) = 0.5$$

Bayes' theorem

$$P(h|e) = \frac{P(h)*P(e|h)}{P(e)}$$

With this information in hand, figuring out the probability of the non-design hypothesis given our evidence is a matter of just plugging in the numbers.

$$\boxed{Pr(\text{non-design}) = 0.5}$$

$$\boxed{Pr(\text{design}) = 0.5}$$

$$\boxed{Pr(e) = 0.25}$$

It is difficult to think about numbers as large as the denominator of this fraction. But to give you some idea: the odds of winning Powerball are about 1 in 300 million. Now consider the odds of winning Powerball one trillion times in a row. Call that a "super Powerball."

Now consider the odds of winning a super Powerball one trillion times in a row. Call that a "super duper Powerball."

Now consider the odds of winning a super duper Powerball one trillion times in a row. The odds of this happening are about $1 / 10^{44}$ — so much, much higher than the odds of the cosmological constant falling in the life-permitting range by chance.

This means that if you simply take the physics at face value, and begin by assigning a probability of 0.5 to the non-design hypothesis, you should think that the chances of the non-design hypothesis being true are vastly lower than the chances of winning a super duper powerball a trillion times in a row.

fine-
tuning of the
universe

Bayes'
theorem

the argument

objections

This is the probability of the non-design hypothesis being true, given that the cosmological constant is in the life-permitting range.

$$Pr(\text{non-design} \mid \text{life}) \approx \frac{2}{10^{120}}$$

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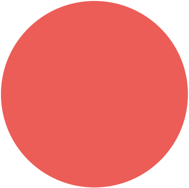
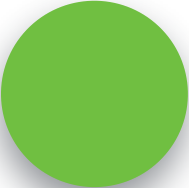
fine-tuning of the universe





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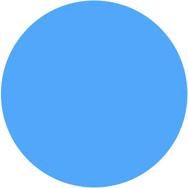

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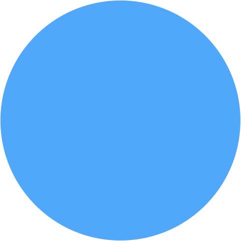

$$Pr(\text{non-design} \mid e) = \frac{0.5 * \frac{1}{10^{120}}}{0.25} = \frac{2}{10^{120}}$$

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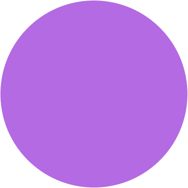
minus the very small number we were just discussing:

By contrast, the probability of the design hypothesis is very close to 1. It is, in fact, 1

$$Pr(\text{design} \mid e) = 1 - \frac{2}{10^{120}}$$

This is about as close to certainty as it is possible to get.



3. P(design)=P(non-design)=0.5.

intelligent designer, then God exists.

2. P

5. P

8. If the universe was created by an

4. Bayes' theorem.

1. P

. (1,2,3,4)

6. Life exists.

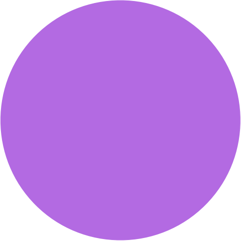
C. $P(\text{God exists}) \approx 1$. (7,8)

7. P(design) $\approx$ 1. (5,6)

$$(\texttt{life}|\texttt{design}) = 0.5$$

$$(\texttt{life}|\texttt{non-design}) = \frac{1}{10^{120}}$$

$$(\texttt{design}|\texttt{life}) \approx 1$$



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